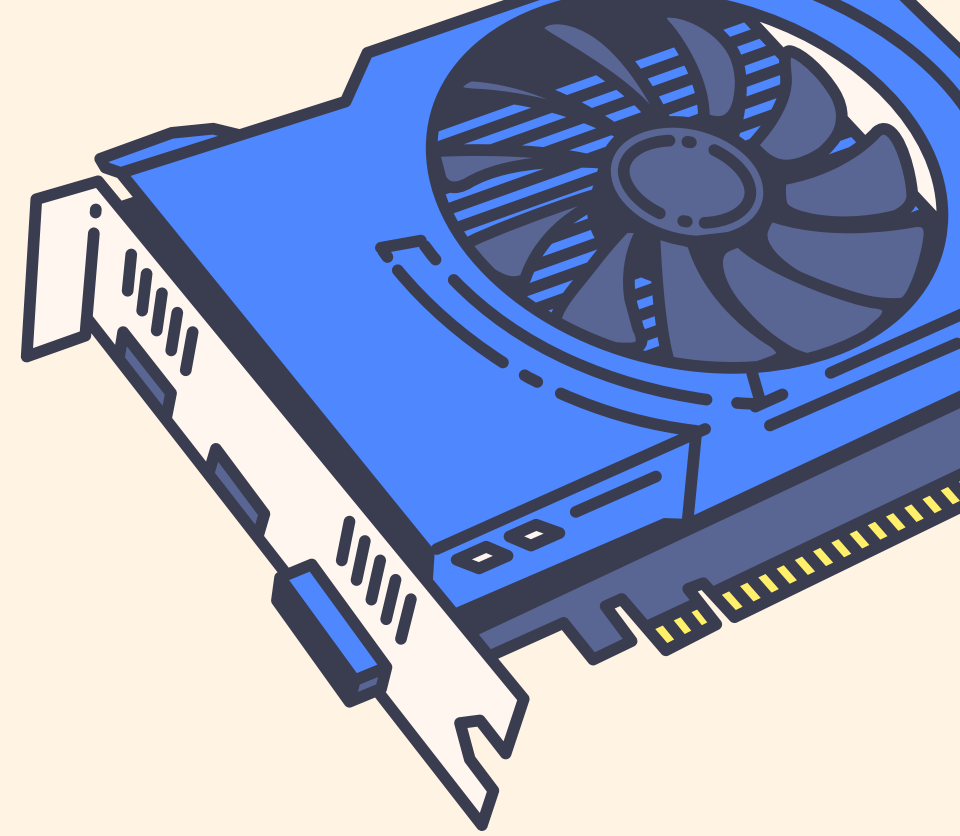
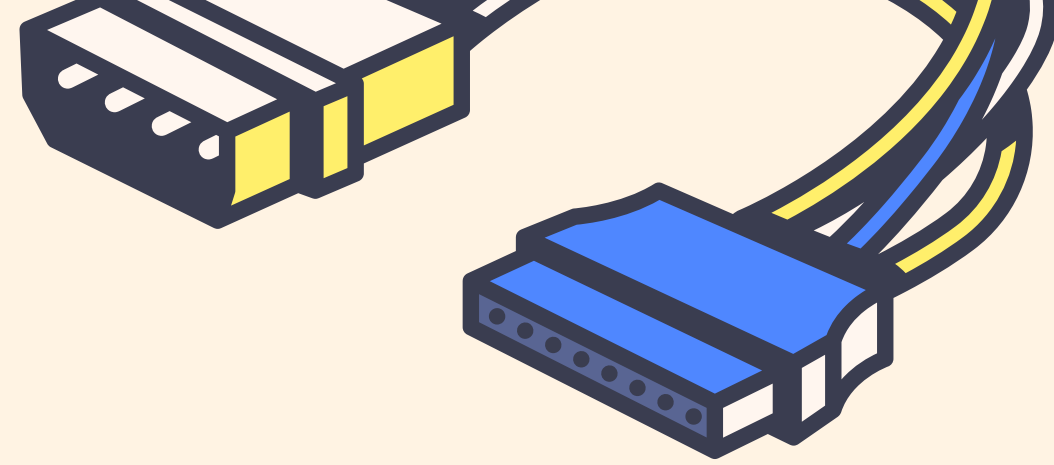
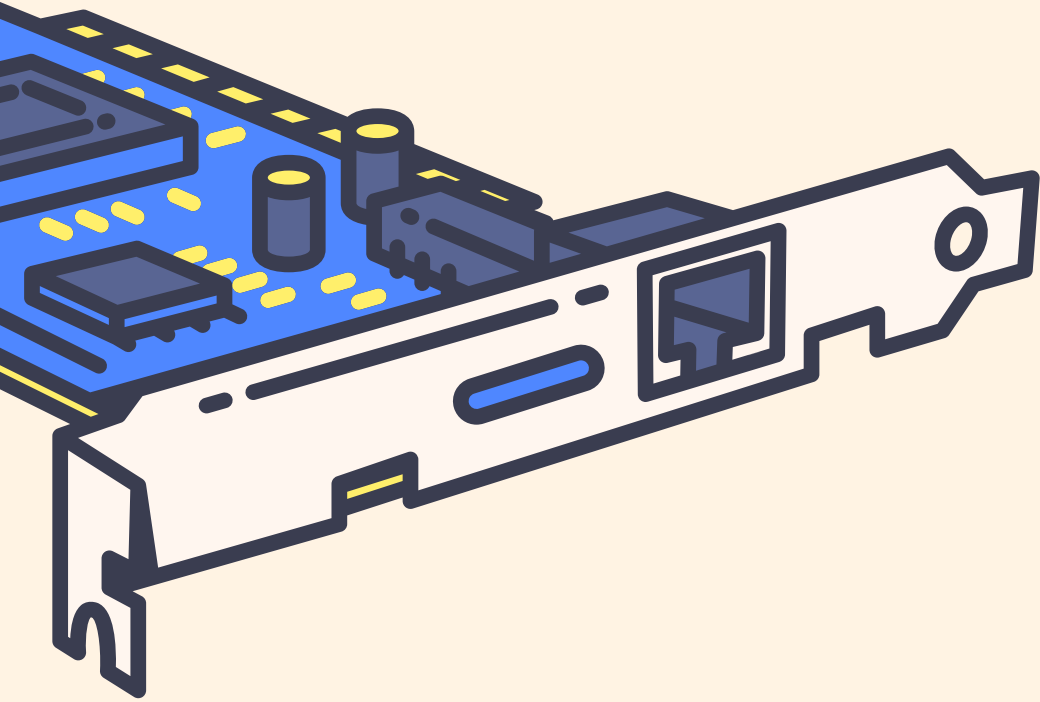
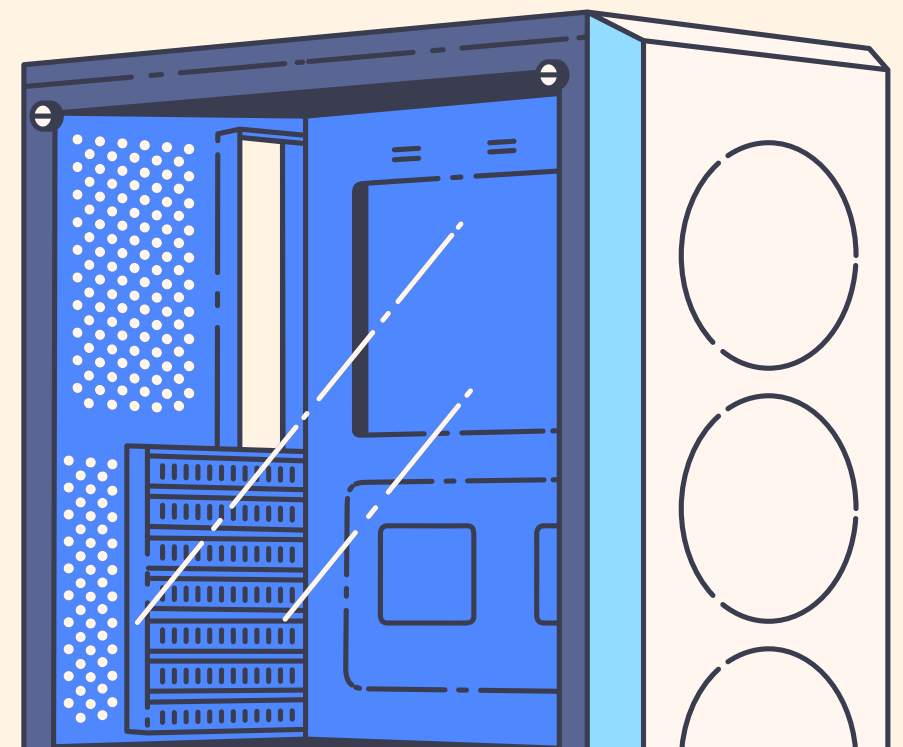
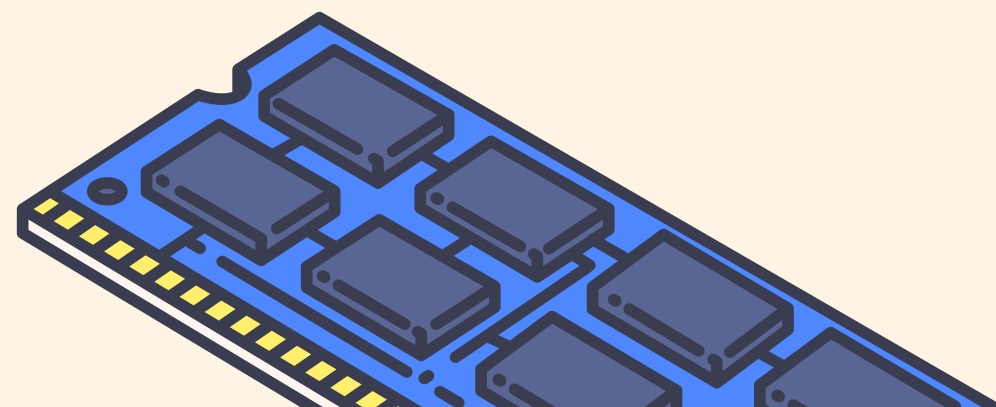
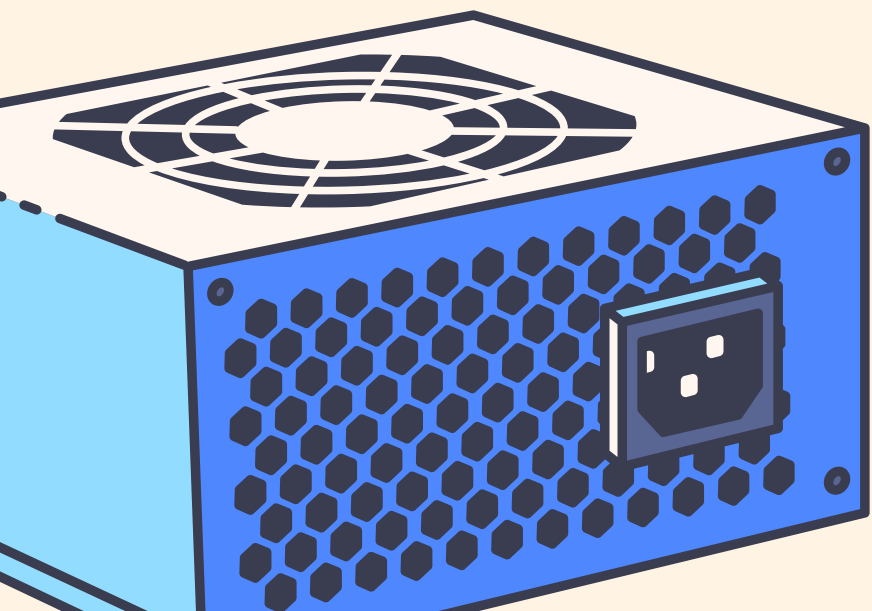
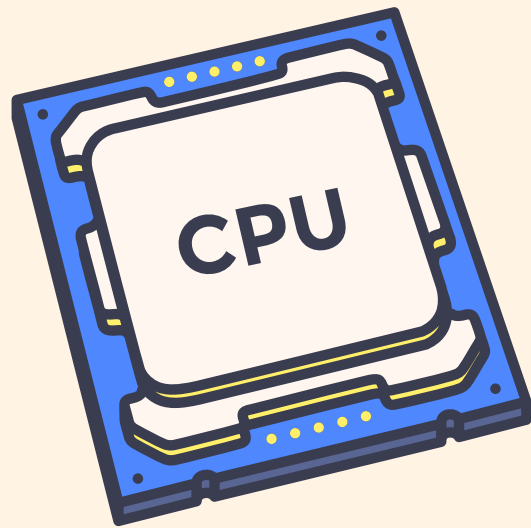
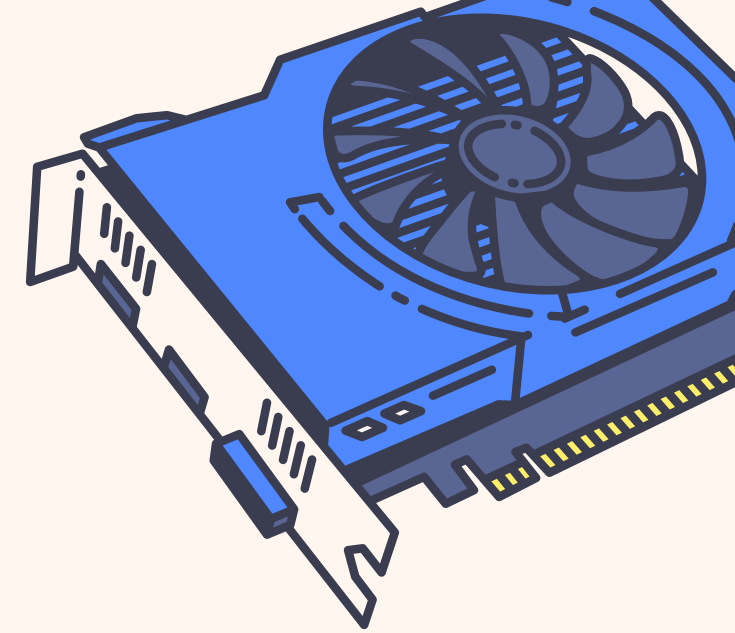
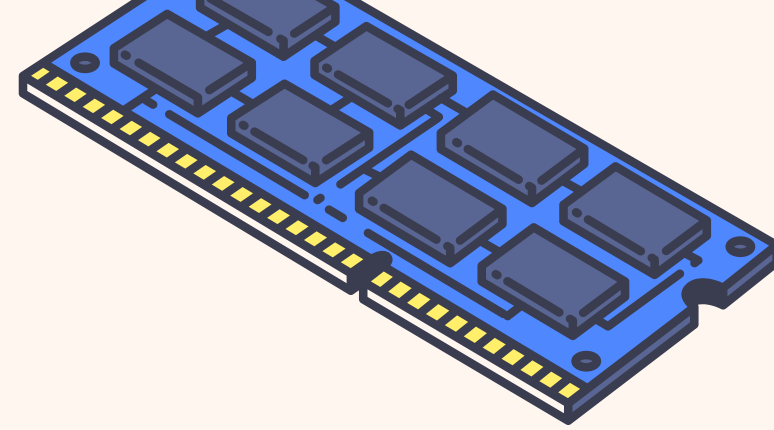
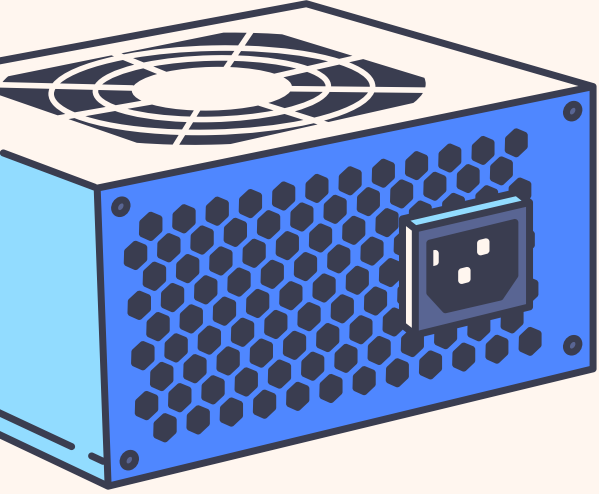




AtliQ Hardware

Project Presentation





Introduction

Our Company

Atliq Hardwares – One of the leading computer hardware producers in India.

Background

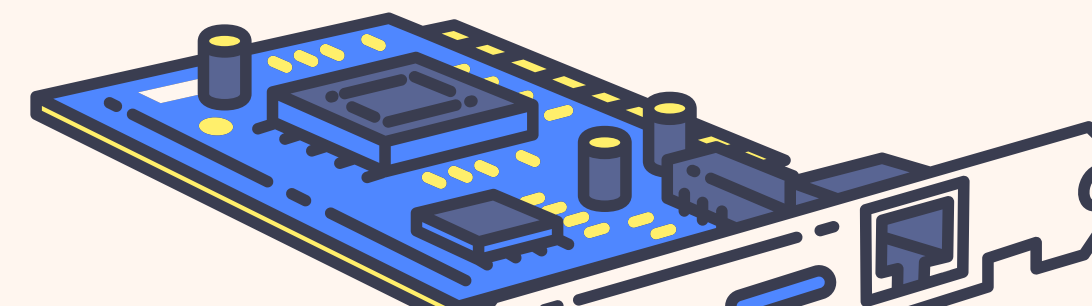
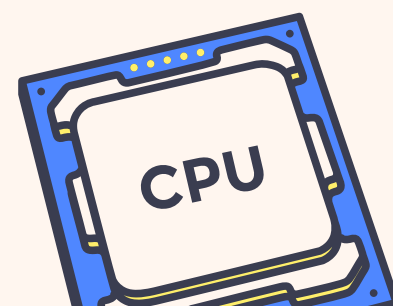
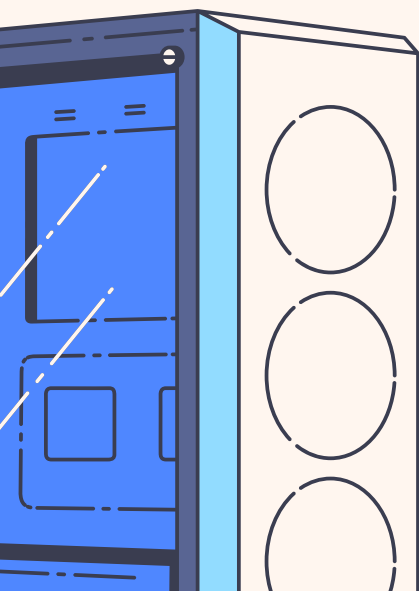
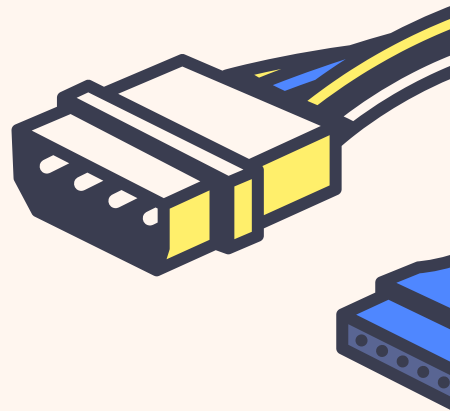
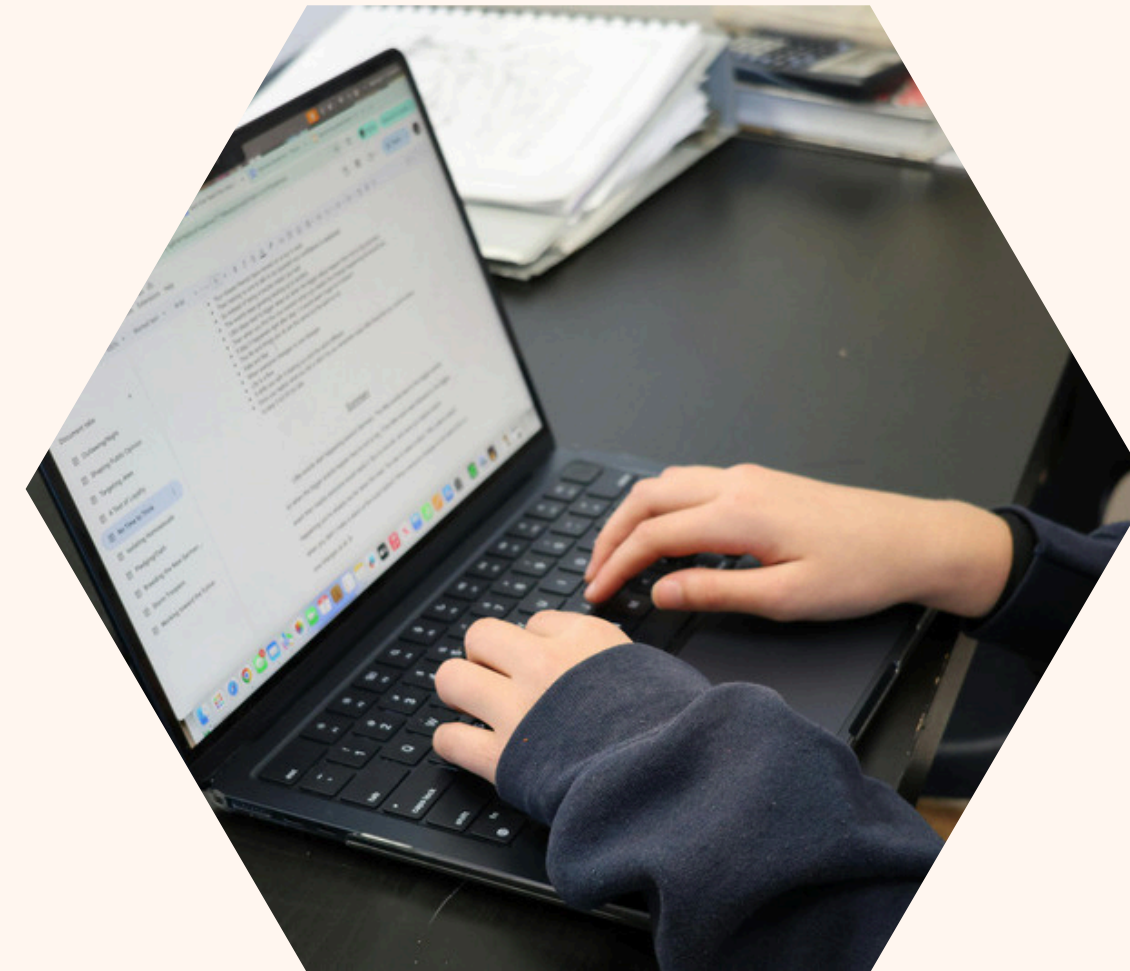
The management noticed that they do not get enough insights to make quick and smart data-informed decisions.

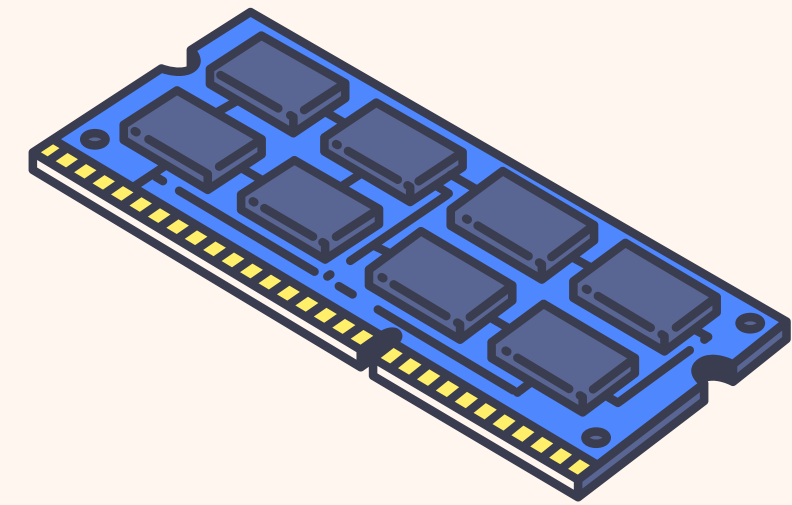
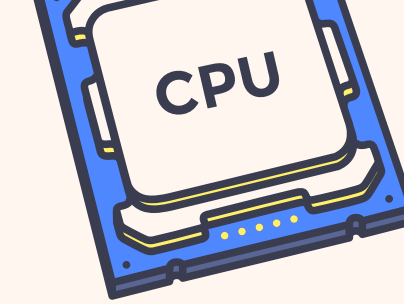
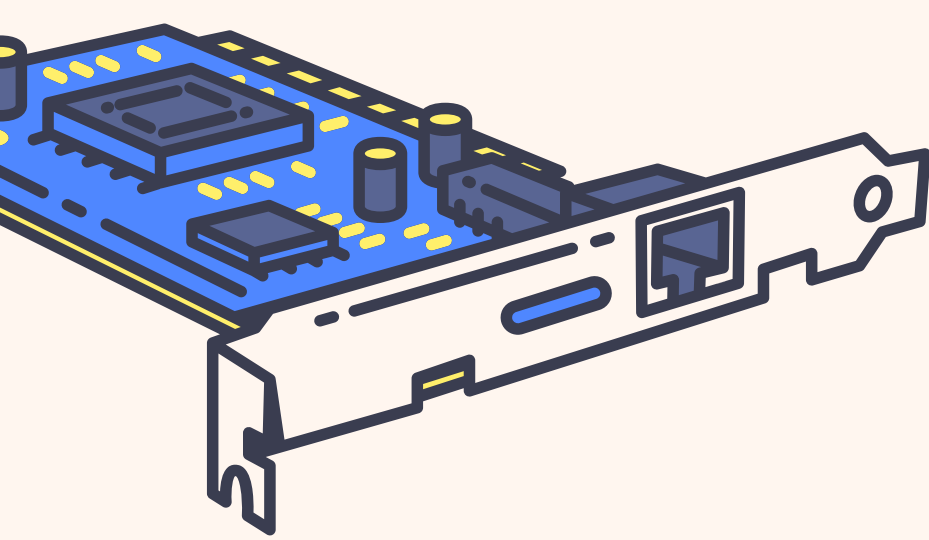
Problem

There are 10 ad-hoc requests for which the company needs insights.

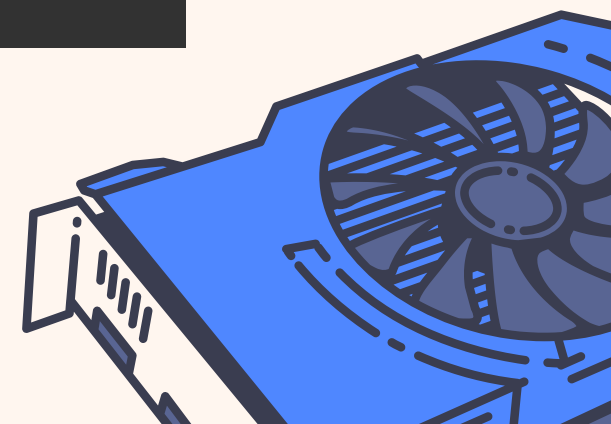
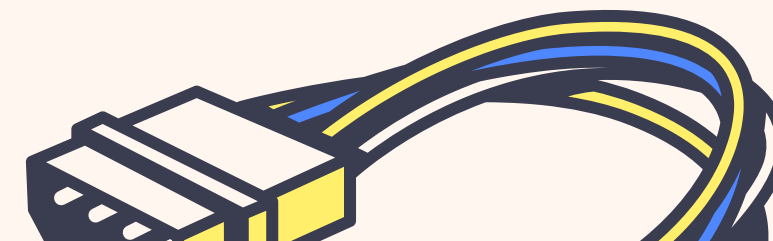
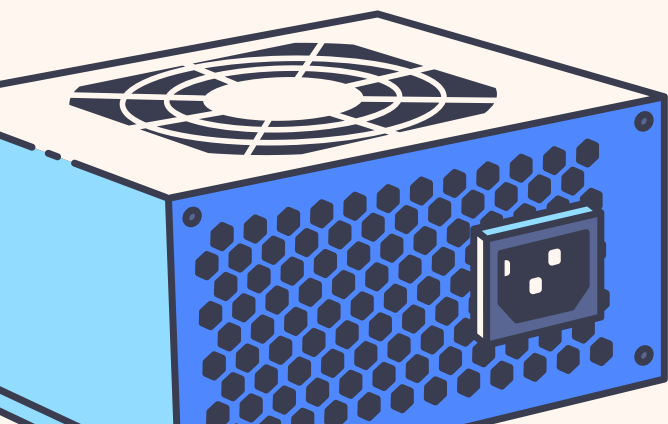
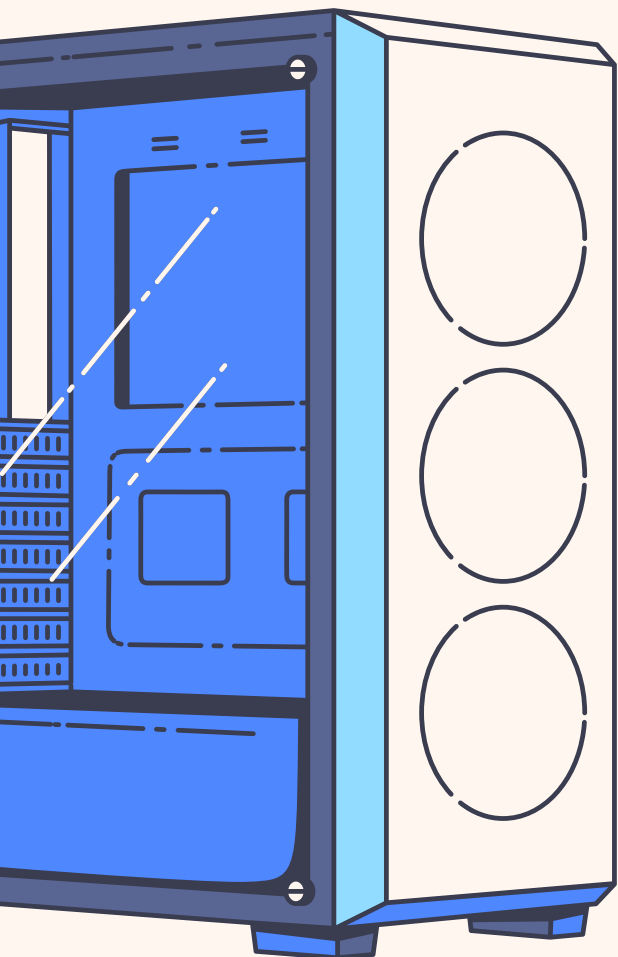
Approach

Run a SQL query to answer these requests. Convert it into visualizations and present the insights to the top-level management





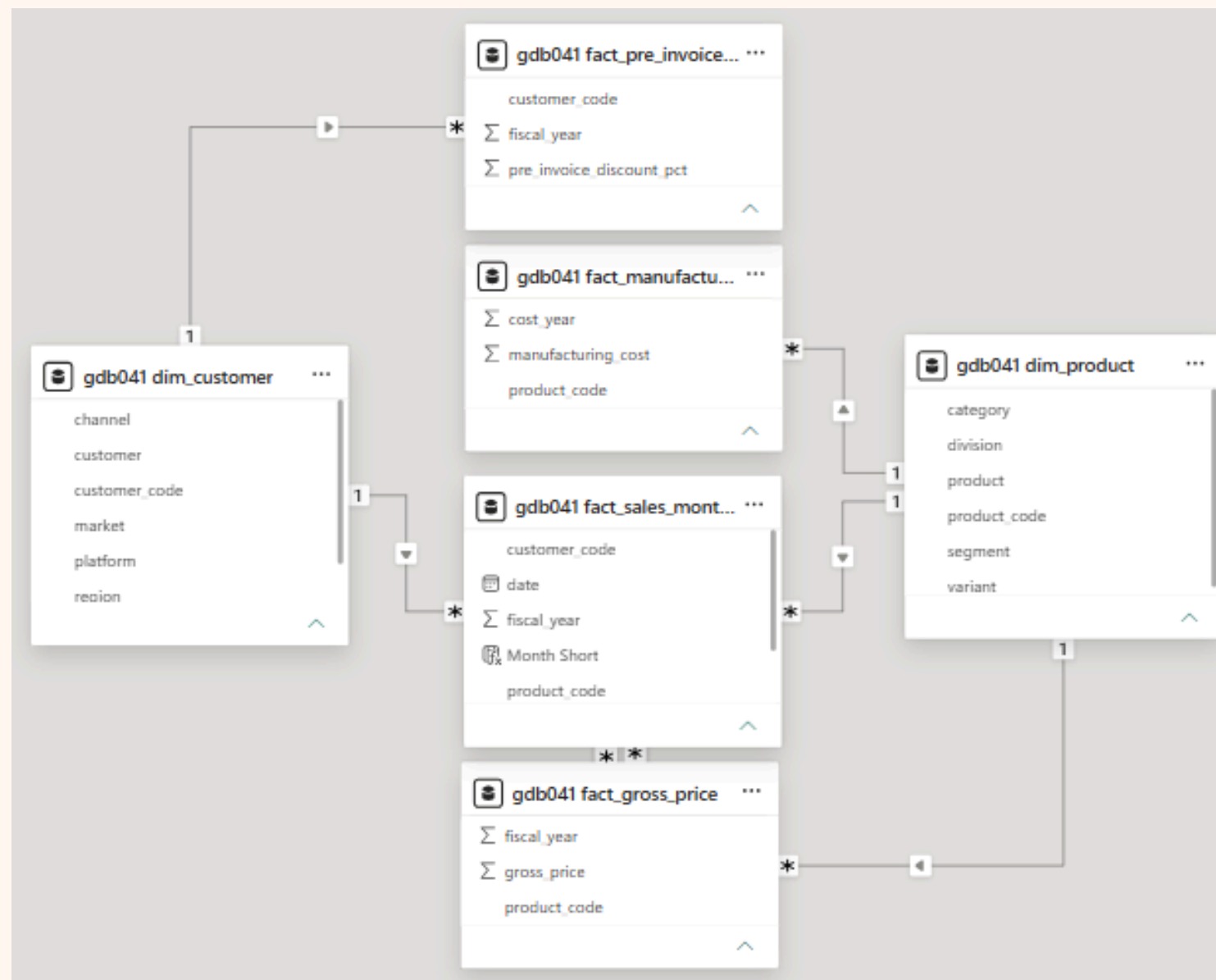
AtliQ's Business Their Markets



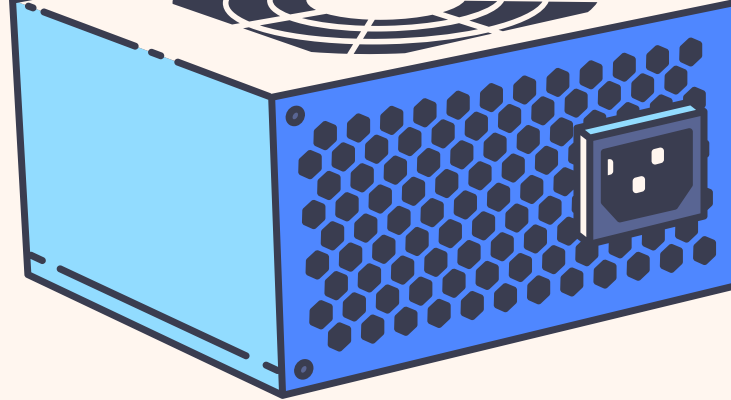
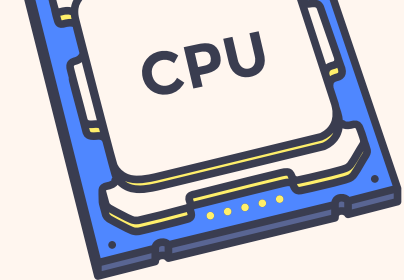
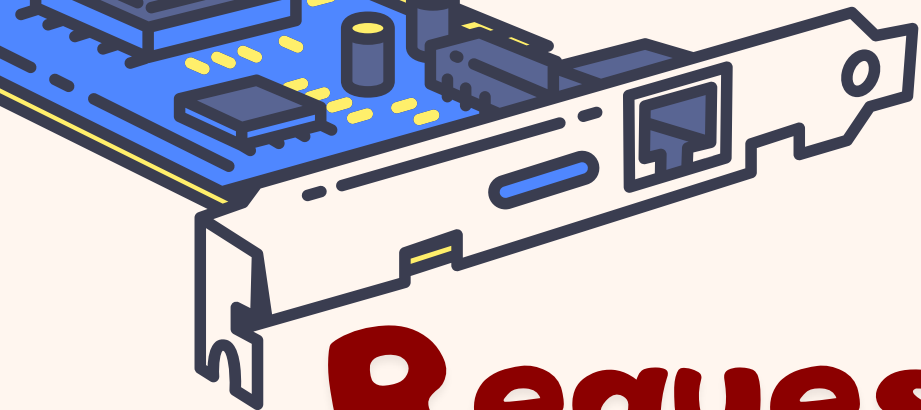
AtliQ's Product Line



Data Modleing



- Input data consists of sales data for FY 2020 and FY 2021, along with different dimension tables like customer details, product details, etc



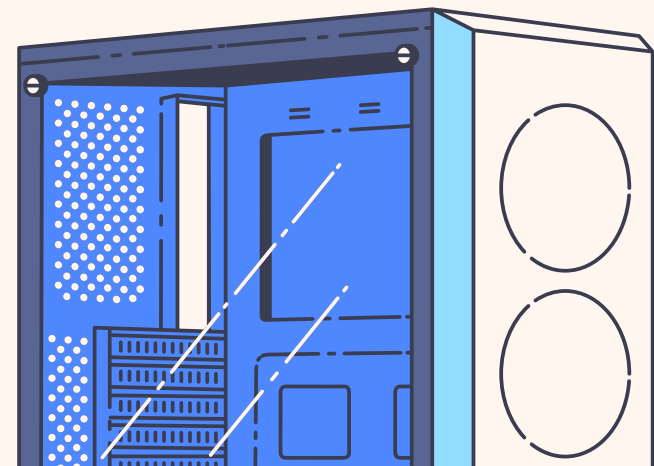
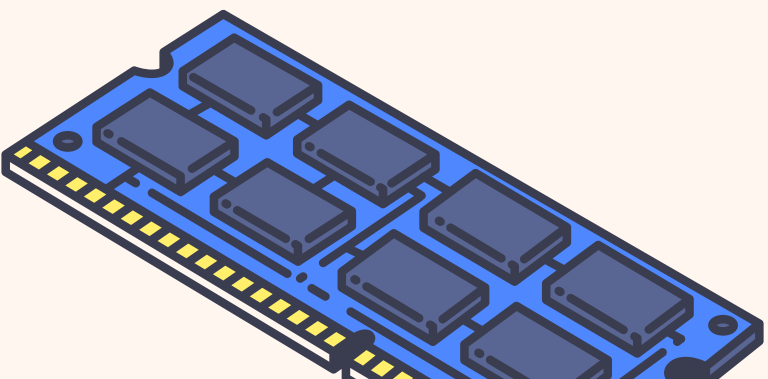
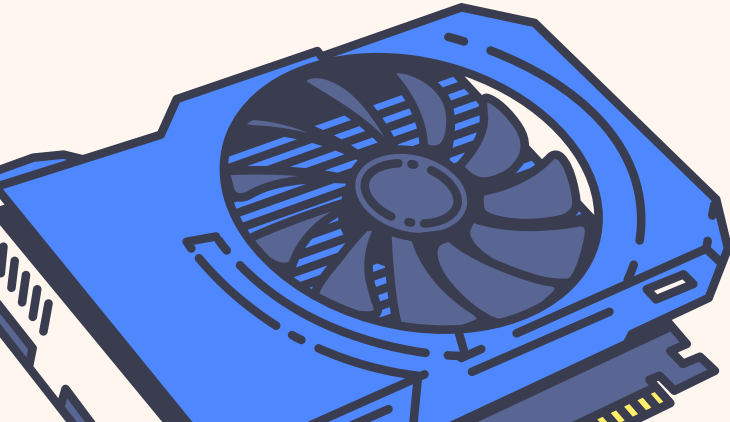
Request 1 :

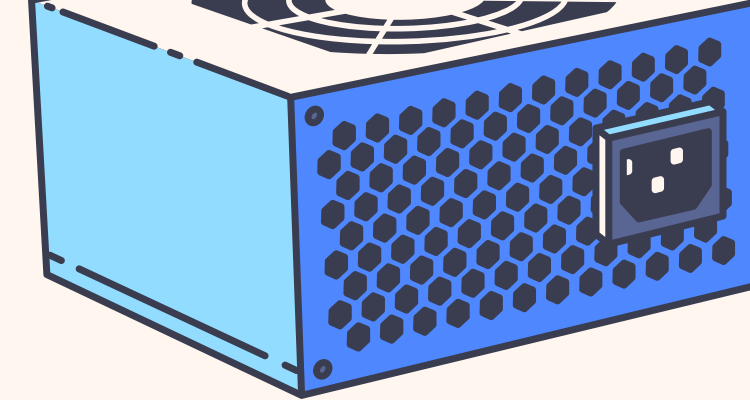
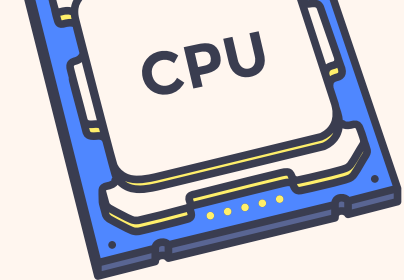
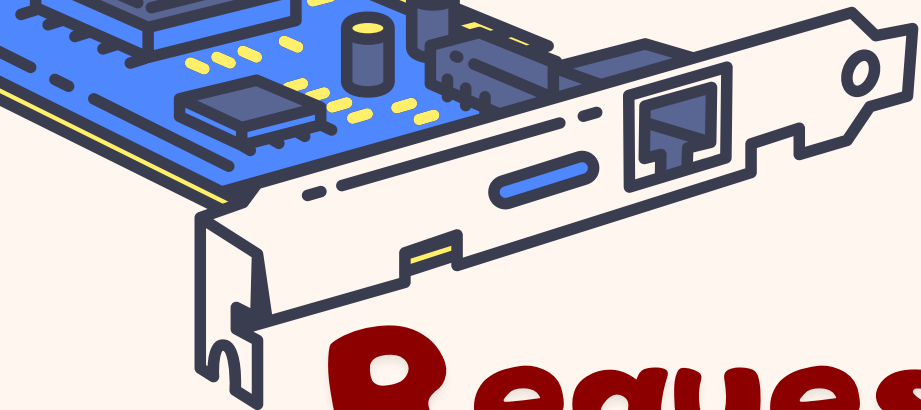
Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.



```
SELECT
  market, customer
FROM
  gdb023.dim_customer
WHERE
  customer = 'Atliq Exclusive'
  AND region = 'APAC'
;
```

market	customer
India	Atliq Exclusive
Indonesia	Atliq Exclusive
Japan	Atliq Exclusive
Philippines	Atliq Exclusive
South Korea	Atliq Exclusive
Australia	Atliq Exclusive
Newzealand	Atliq Exclusive
Bangladesh	Atliq Exclusive



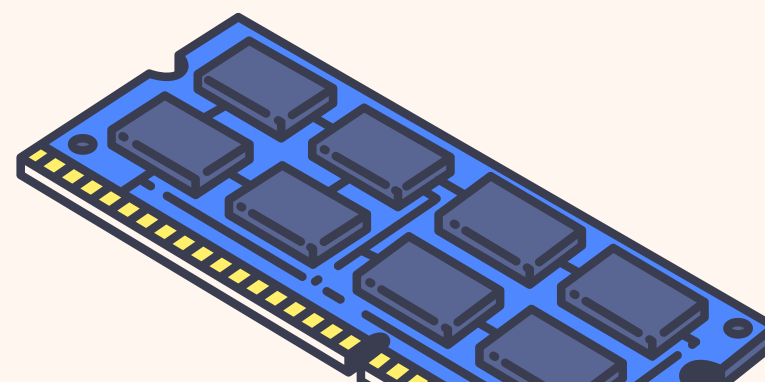
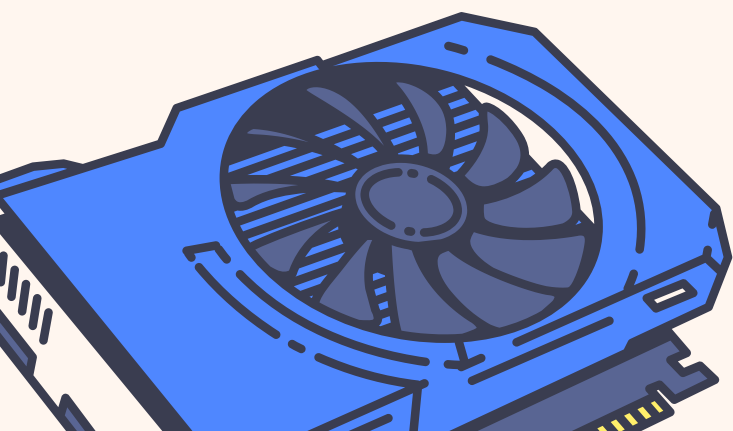


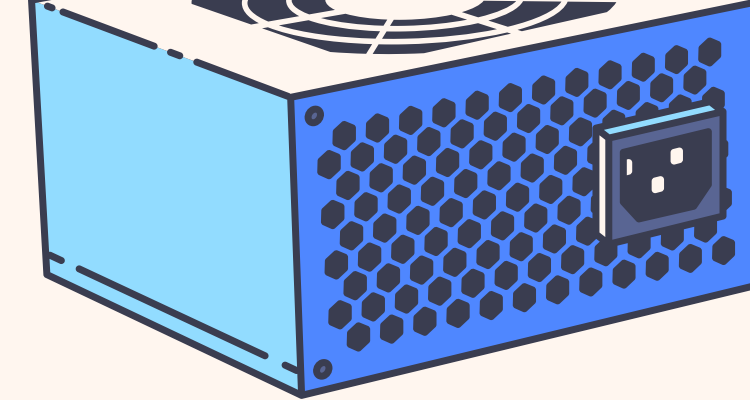
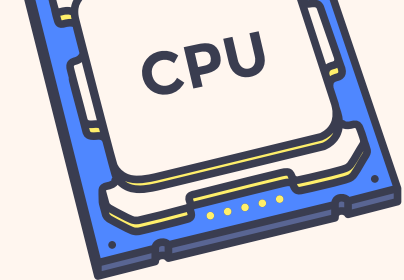
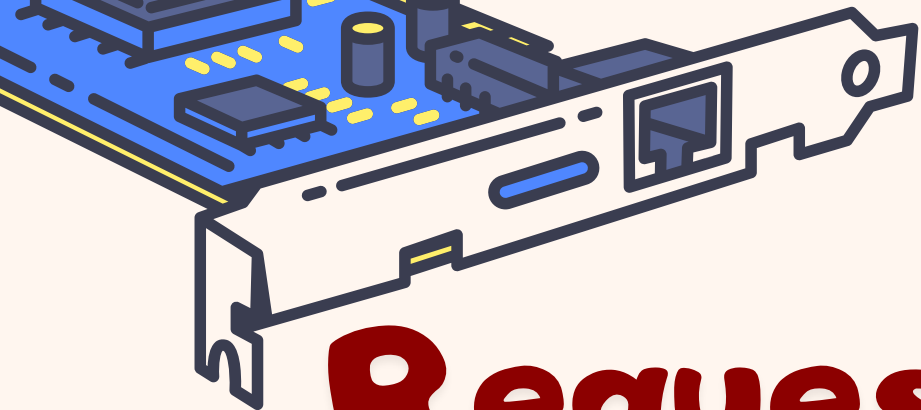
Request 2 :

What is the percentage of unique product increase in 2021 vs. 2020?
The final output contains these fields

```
SELECT
  COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2020 THEN p.product_code
  END) AS product_code_2020,
  COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2021 THEN p.product_code
  END) AS product_code_2021,
  ((COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2021 THEN p.product_code
  END) - COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2020 THEN p.product_code
  END)) / COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2020 THEN p.product_code
  END)) * 100 AS pct
FROM
  fact_sales_monthly s
  JOIN
  dim_product p ON p.product_code = s.product_code
```

product_code_2020	product_code_2021	pct
245	334	36.3265





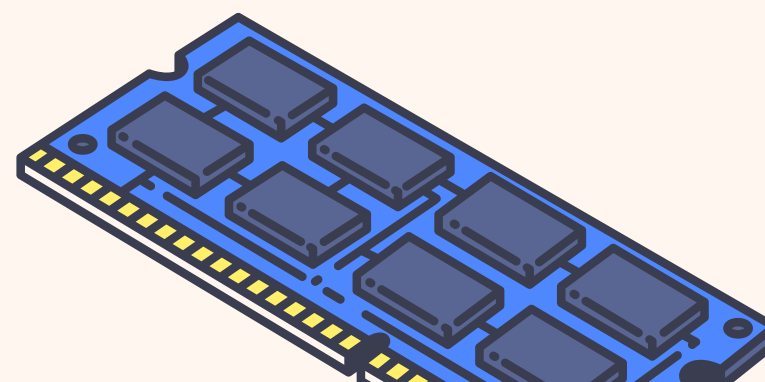
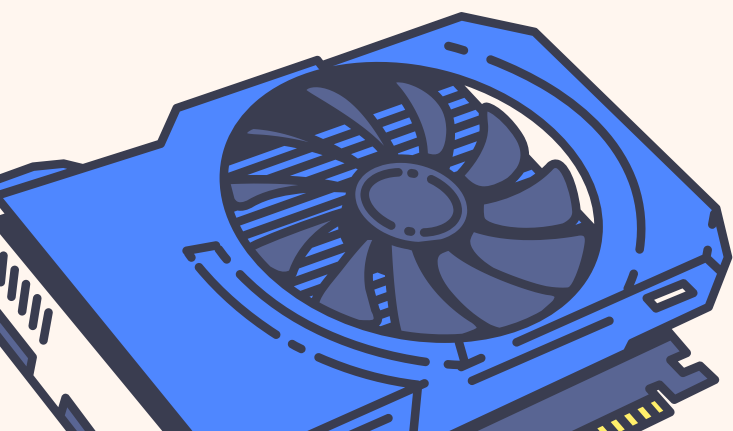
Request 3 :

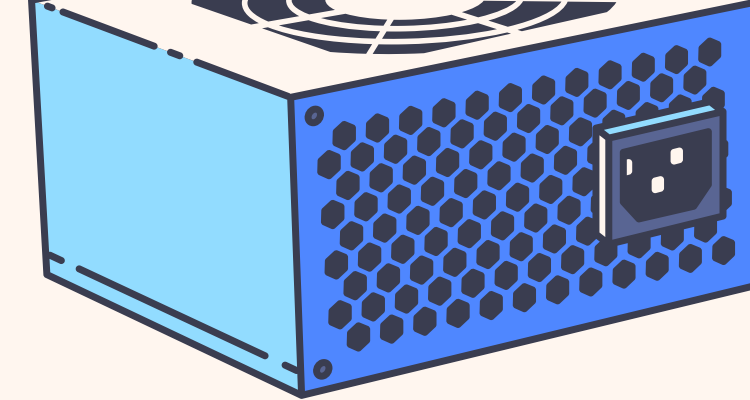
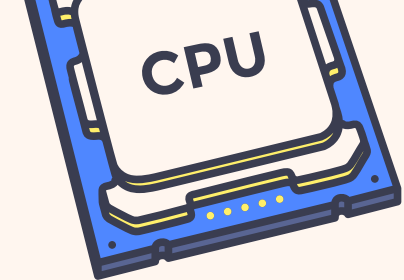
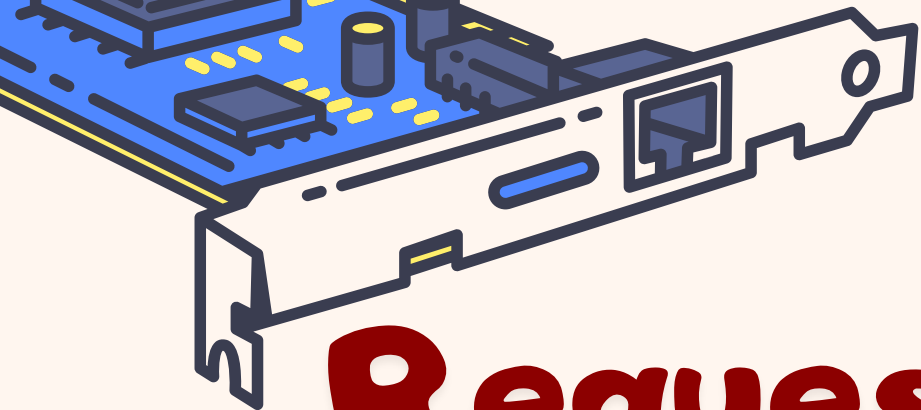
Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.



```
SELECT
    segment, COUNT(DISTINCT (product_code)) AS product_counts
FROM
    dim_product
GROUP BY segment
ORDER BY product_counts DESC;
```

segment	product_counts
Notebook	129
Accessories	116
Peripherals	84
Desktop	32
Storage	27
Networking	9





Request 4 :

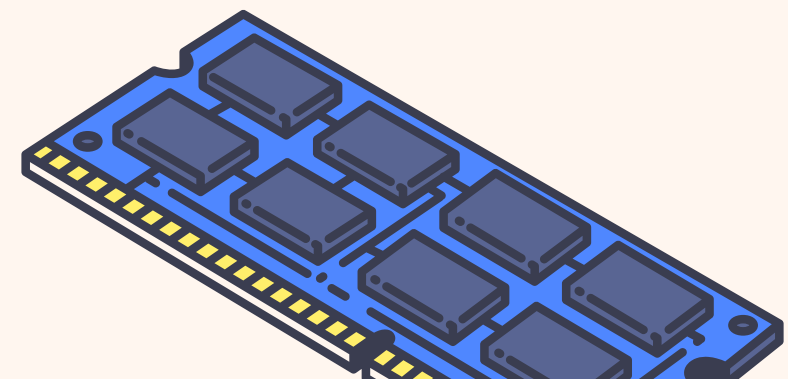
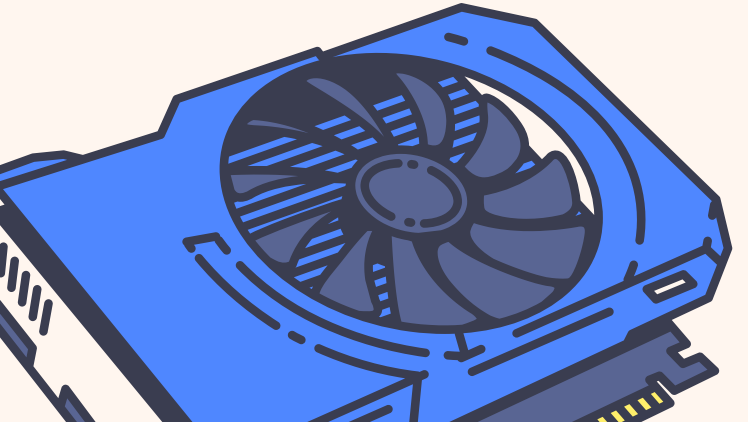
Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

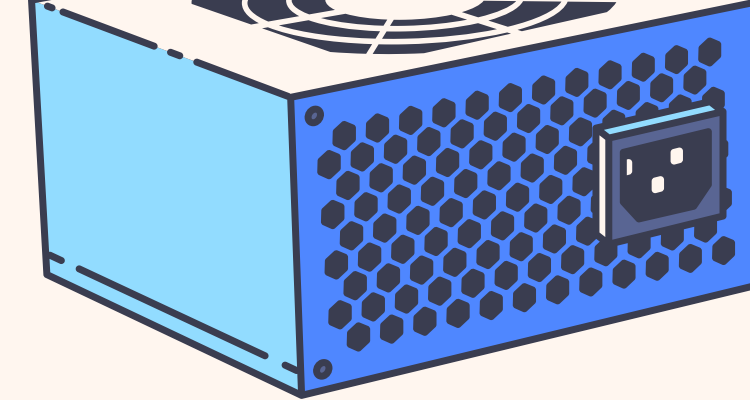
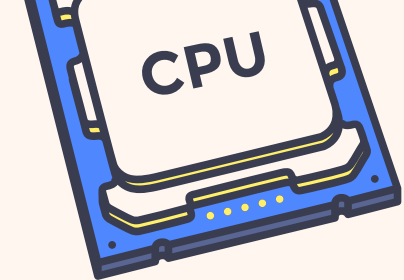
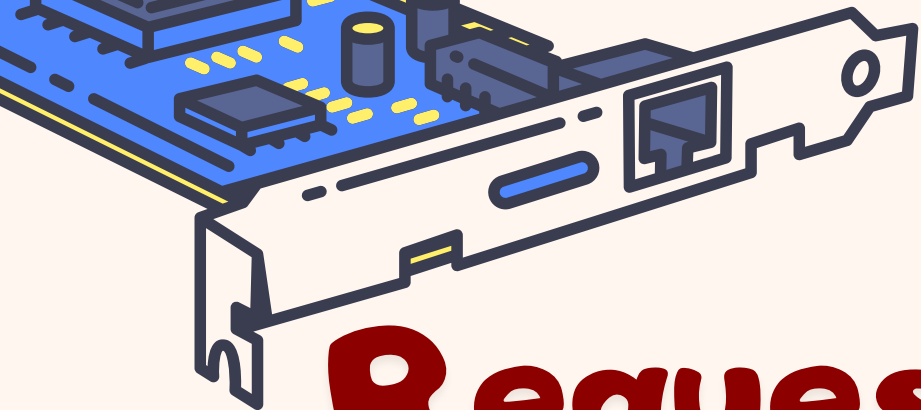


```
SELECT
  p.segment,
  COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2020 THEN p.product_code
  END) AS product_code_2020,
  COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2021 THEN p.product_code
  END) AS product_code_2021,
  COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2021 THEN p.product_code
  END) - COUNT(DISTINCT CASE
    WHEN s.fiscal_year = 2020 THEN p.product_code
  END) AS difference
FROM
  fact_sales_monthly s
  JOIN
  dim_product p ON p.product_code = s.product_code
GROUP BY p.segment
ORDER BY difference DESC
```

segment	product_code_2020	product_code_2021	difference
Accessories	69	103	34
Notebook	92	108	16
Peripherals	59	75	16
Desktop	7	22	15
Storage	12	17	5
Networking	6	9	3

Segment	Product Count 2020	Product Count 2021	Difference
Accessories	69	103	34 ↑
Notebook	92	108	16 ↑
Peripherals	59	75	16 ↑
Desktop	7	22	15 ↑
Storage	12	17	5 ↑
Networking	6	9	3 ↑



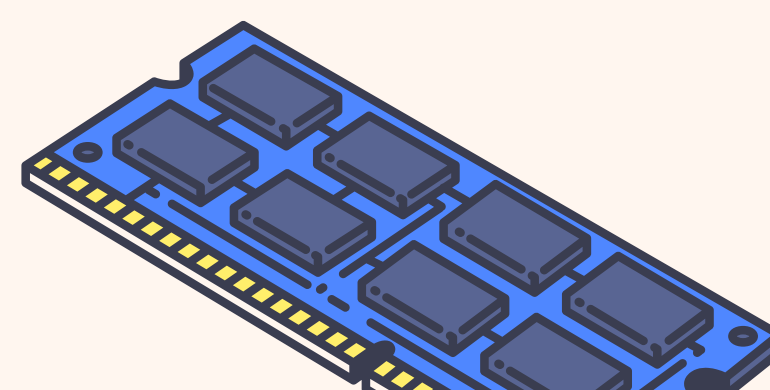
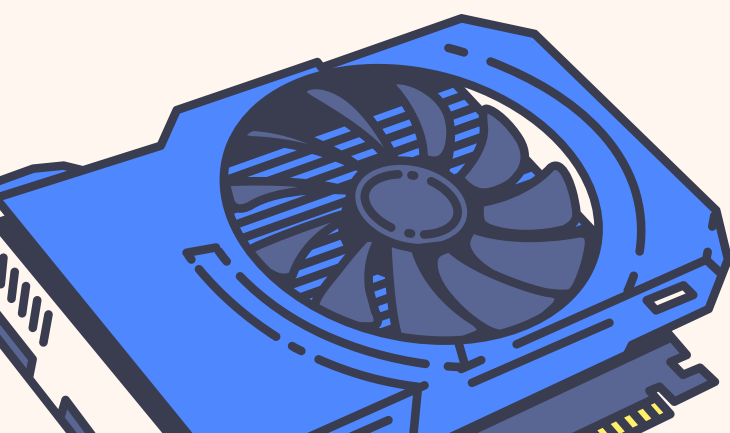


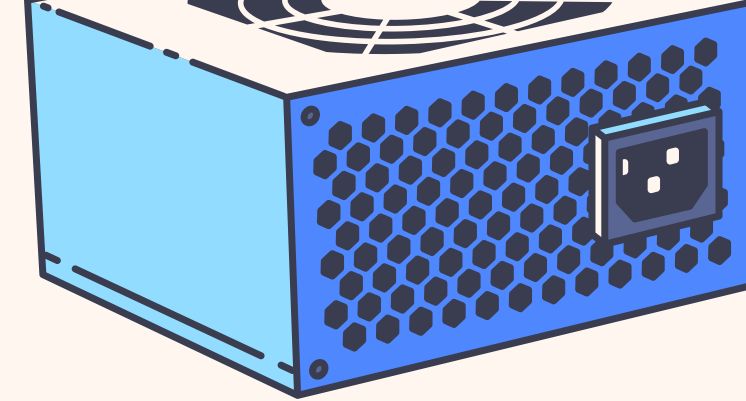
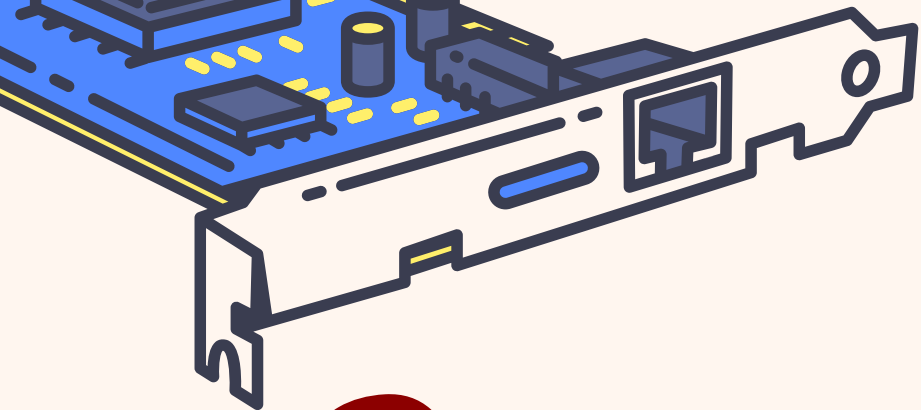
Request 5 :

Get the products that have the highest and lowest manufacturing costs.

```
SELECT F.product_code, P.product, F.manufacturing_cost
FROM fact_manufacturing_cost F JOIN dim_product P
ON F.product_code = P.product_code
WHERE manufacturing_cost
IN (
    SELECT MAX(manufacturing_cost) FROM fact_manufacturing_cost
    UNION
    SELECT MIN(manufacturing_cost) FROM fact_manufacturing_cost
)
ORDER BY manufacturing_cost DESC ;
```

product_code	product	manufacturing_cost
A6120110206	AQ HOME Allin1 Gen 2	240.5364
A2118150101	AQ Master wired x1 Ms	0.8920



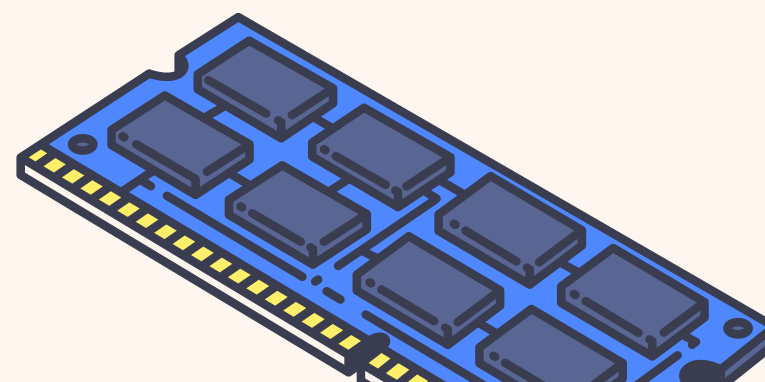
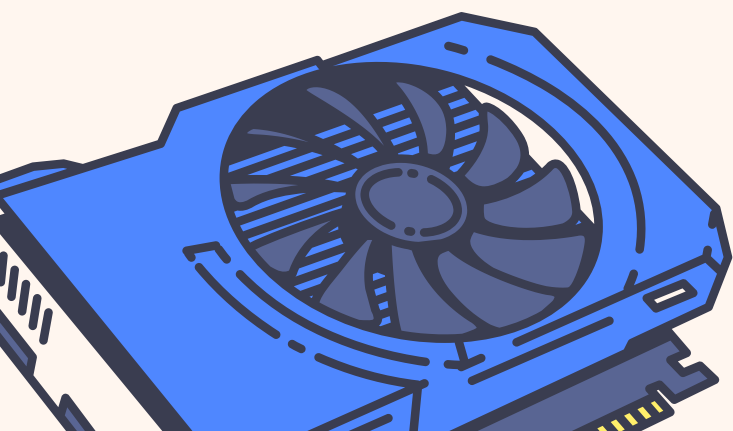


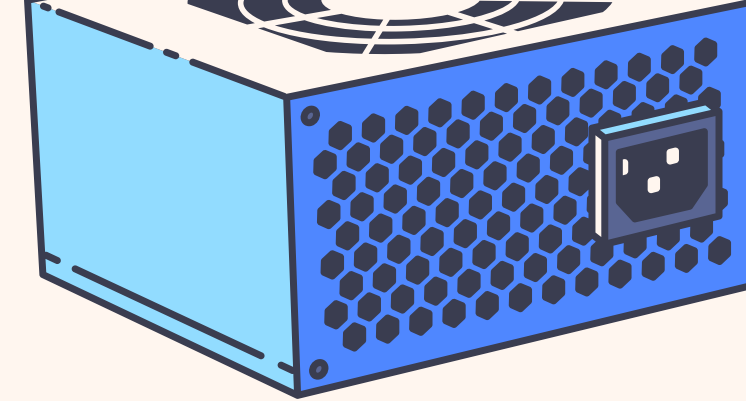
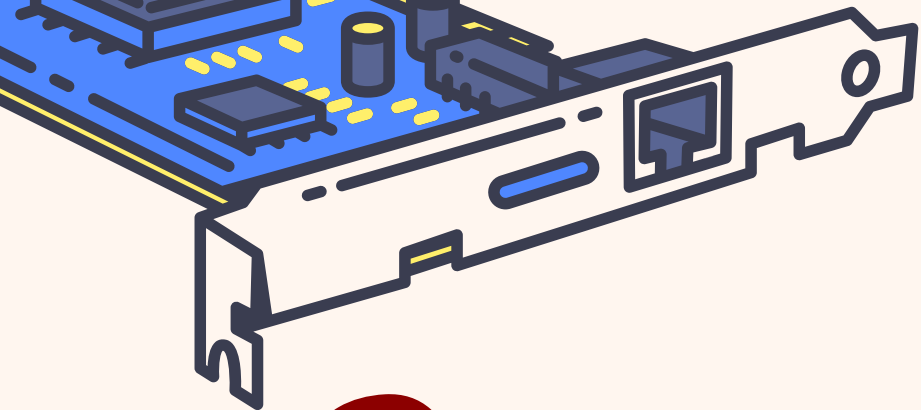
Request 6 :

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

```
SELECT
  c.customer_code, c.customer, p.pre_invoice_discount_pct
FROM
  gdb023.dim_customer c
  JOIN
    fact_pre_invoice_deductions p ON p.customer_code = c.customer_code
WHERE
  pre_invoice_discount_pct > (SELECT
    AVG(pre_invoice_discount_pct)
  FROM
    fact_pre_invoice_deductions)
  AND c.market = 'india'
ORDER BY pre_invoice_discount_pct DESC
LIMIT 5
```

customer_code	customer	pre_invoice_discount_pct
90002009	Flipkart	0.3083
90002006	Viveks	0.3038
90002003	Ezone	0.3028
90002002	Croma	0.3025
90002016	Amazon	0.2933



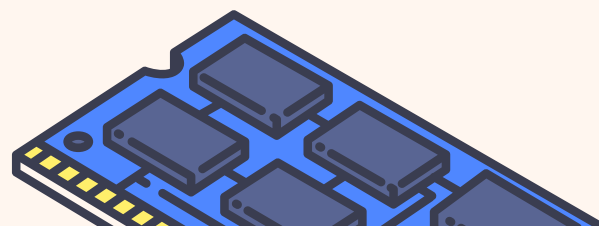
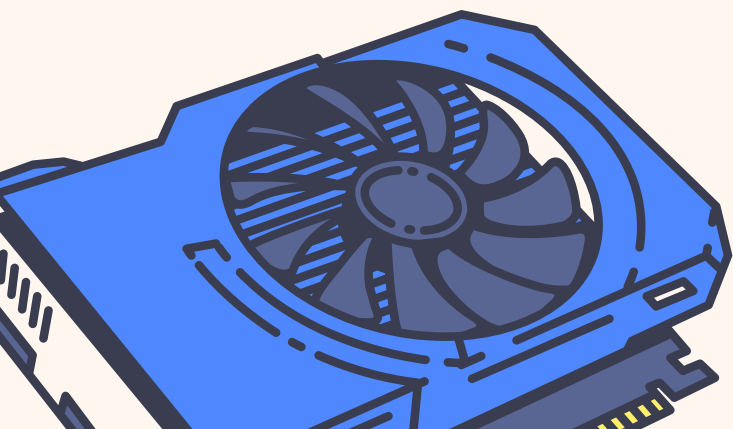


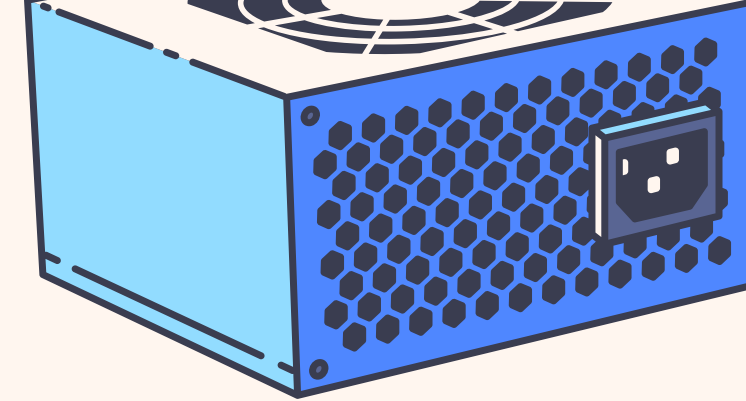
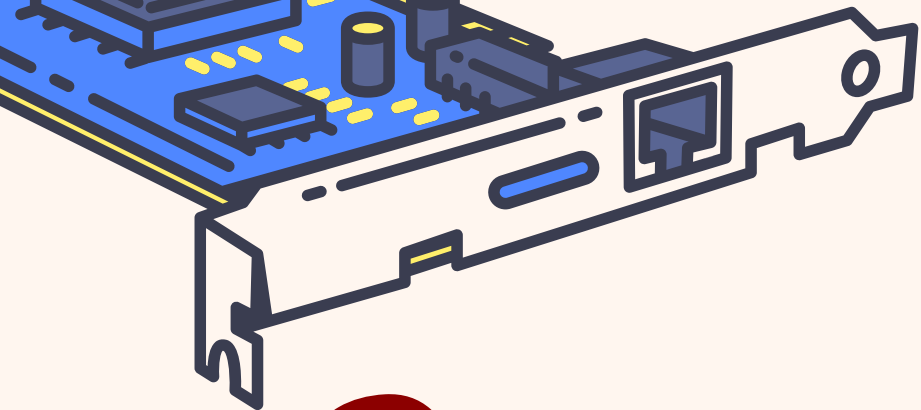
Request 7 :

Get the complete report of the Gross sales amount for the customer “Atliq Exclusive” for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions.

```
SELECT
  MONTH(date) AS month,
  MONTHNAME(date) AS month_name,
  s.fiscal_year,
  ROUND(SUM(s.sold_quantity * g.gross_price), 2) AS groos_sales
FROM
  gdb023.fact_sales_monthly s
  JOIN
  fact_gross_price g ON g.product_code = s.product_code
  JOIN
  dim_customer c ON c.customer_code = s.customer_code
WHERE
  c.customer = 'Atliq Exclusive'
GROUP BY month , fiscal_year
ORDER BY fiscal_year
```

month	month_name	fiscal_year	groos_sales
9	September	2020	9092670.34
10	October	2020	10378637.60
11	November	2020	15231894.97
12	December	2020	9755795.06
1	January	2020	9584951.94
2	February	2020	8083995.55
3	March	2020	766976.45
4	April	2020	800071.95
5	May	2020	1586964.48
6	June	2020	3429736.57
7	July	2020	5151815.40
8	August	2020	5638281.83
9	September	2021	19530271.30
10	October	2021	21016218.21
11	November	2021	32247289.79
12	December	2021	20409063.18
1	January	2021	19570701.71
2	February	2021	15986603.89
3	March	2021	19149624.92
4	April	2021	11483530.30
5	May	2021	19204309.41
6	June	2021	15457579.66
7	July	2021	19044968.82
8	August	2021	11324548.34





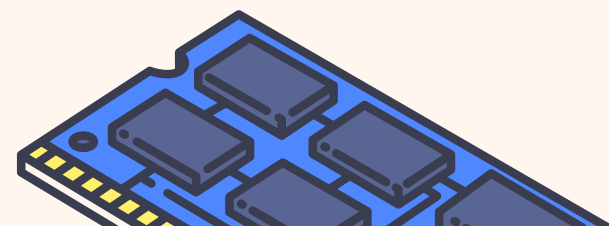
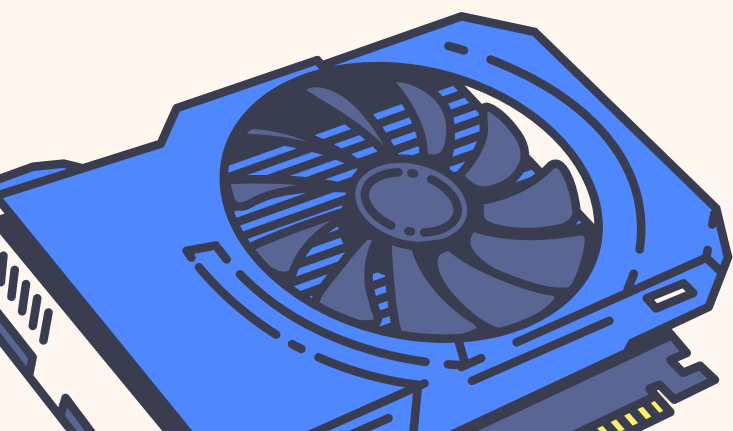
Request 8 :

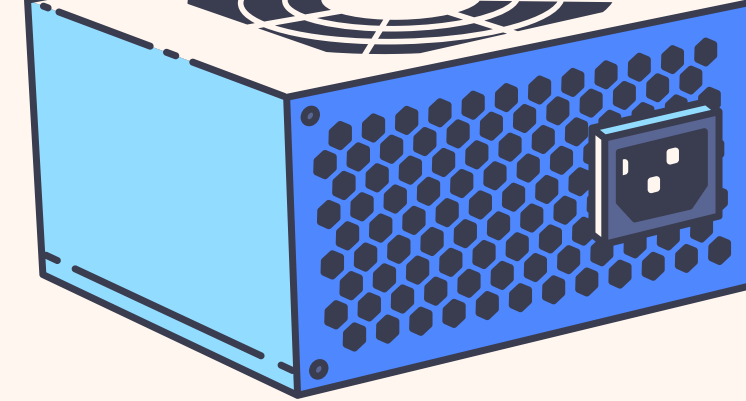
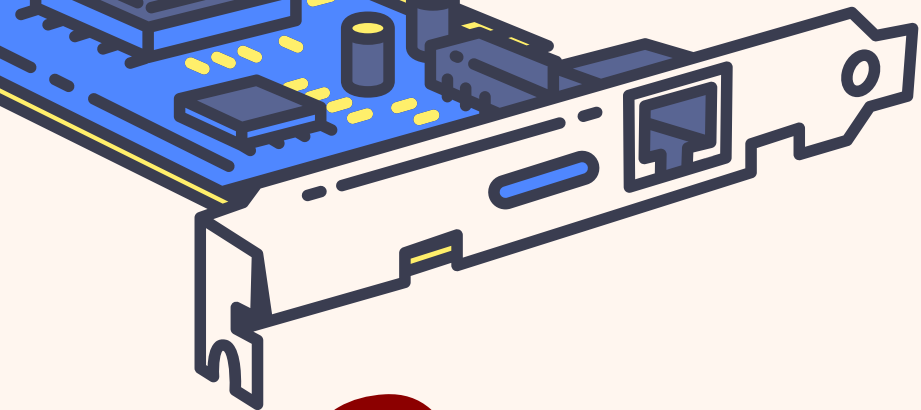
In which quarter of 2020, got the maximum total_sold_quantity?

The final output contains these fields sorted by the total_sold_quantity, Quarter
total_sold_quantity

```
SELECT
CASE
    WHEN MONTH(s.date) IN (9 , 10, 11) THEN 'Q1'
    WHEN MONTH(s.date) IN (12 , 1, 2) THEN 'Q2'
    WHEN MONTH(s.date) IN (3 , 4, 5) THEN 'Q3'
    WHEN MONTH(s.date) IN (6 , 7, 8) THEN 'Q4'
END AS fiscal_quarter,
s.fiscal_year,
SUM(s.sold_quantity) AS total_sold_qty
FROM
    gdb023.fact_sales_monthly s
WHERE
    s.fiscal_year = 2020
GROUP BY fiscal_quarter
ORDER BY total_sold_qty DESC
```

fiscal_quarter	fiscal_year	total_sold_qty
Q1	2020	7005619
Q2	2020	6649642
Q4	2020	5042541
Q3	2020	2075087



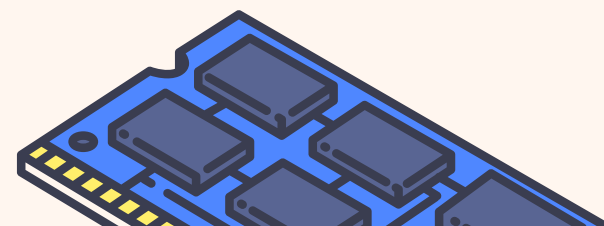
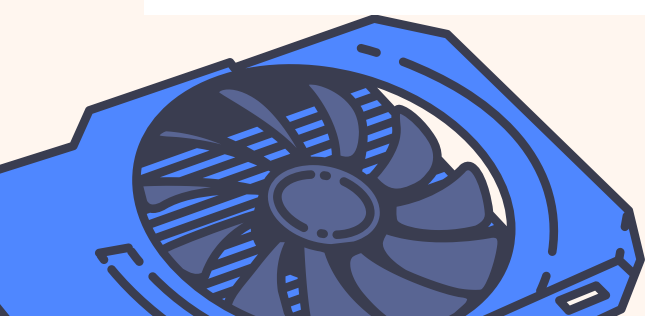
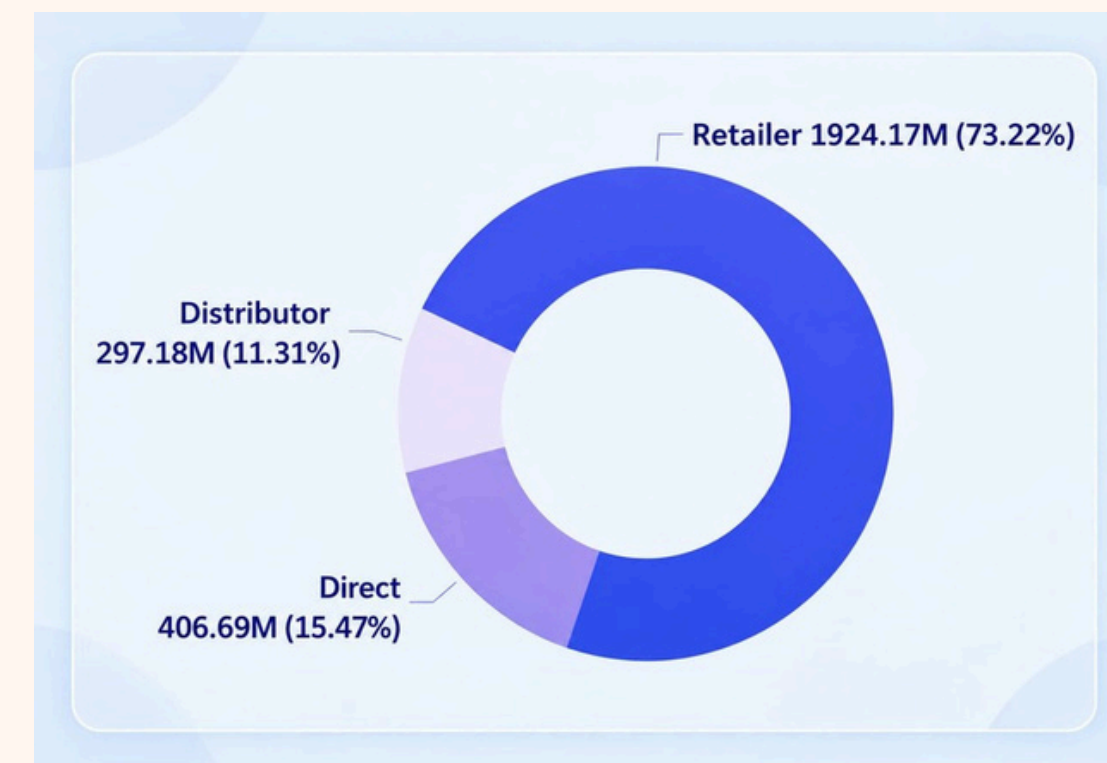


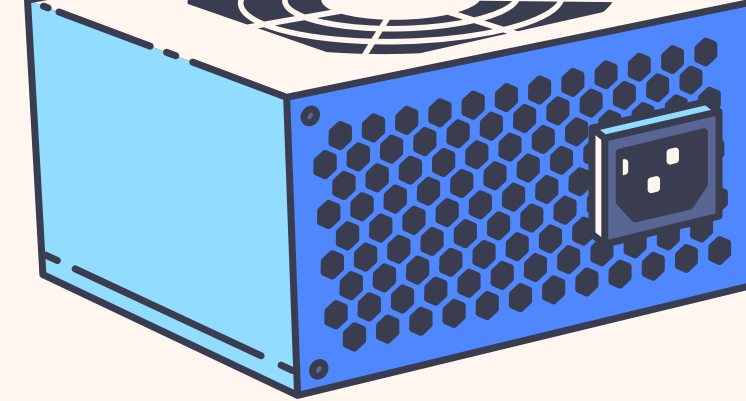
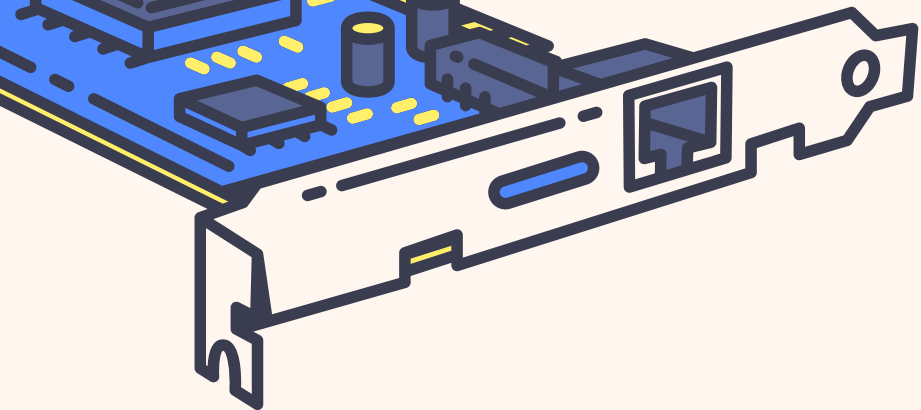
Request 9 :

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

```
With cte1 as(
SELECT
    c.channel,
    ROUND(SUM((s.sold_quantity * g.gross_price) * 1000000),
          2) AS gross_sales_mln
FROM
    gdb023.fact_sales_monthly s
    JOIN
    fact_gross_price g ON g.product_code = s.product_code
    JOIN
    dim_customer c ON c.customer_code = s.customer_code
WHERE
    s.fiscal_year = 2021
GROUP BY channel),
cte2 as(select * ,sum(gross_sales_mln) over() as total_gross_sales_mln
from cte1)
SELECT
    *,
    ROUND((gross_sales_mln / total_gross_sales_mln) * 100,
          2) AS pct_contribution
FROM
    cte2
ORDER BY pct_contribution DESC ;
```

channel	gross_sales_mln	total_gross_sales_mln	pct_contribution
Retailer	1924170397909600.00	2628033151531700.00	73.22
Direct	406686873903300.00	2628033151531700.00	15.47
Distributor	297175879718800.00	2628033151531700.00	11.31



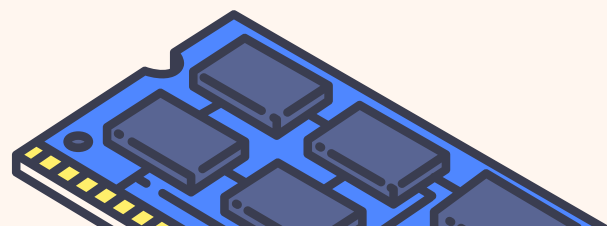
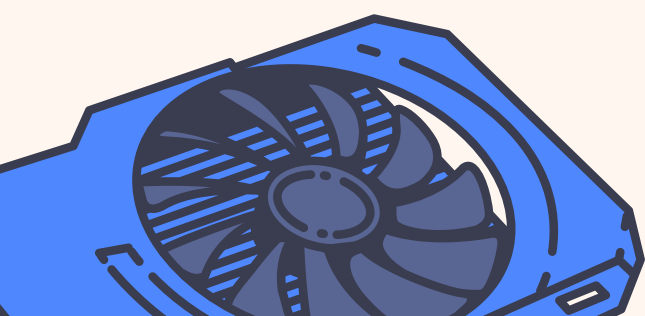


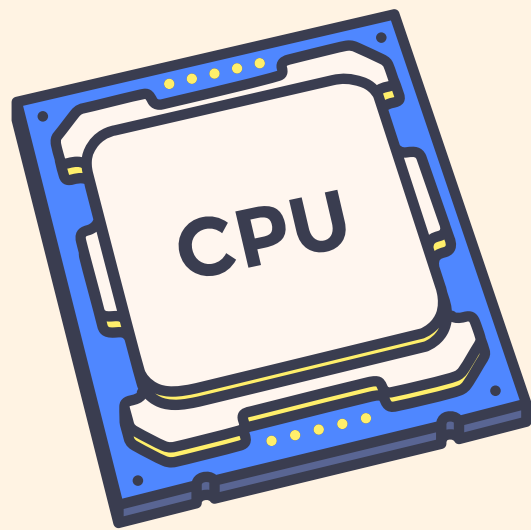
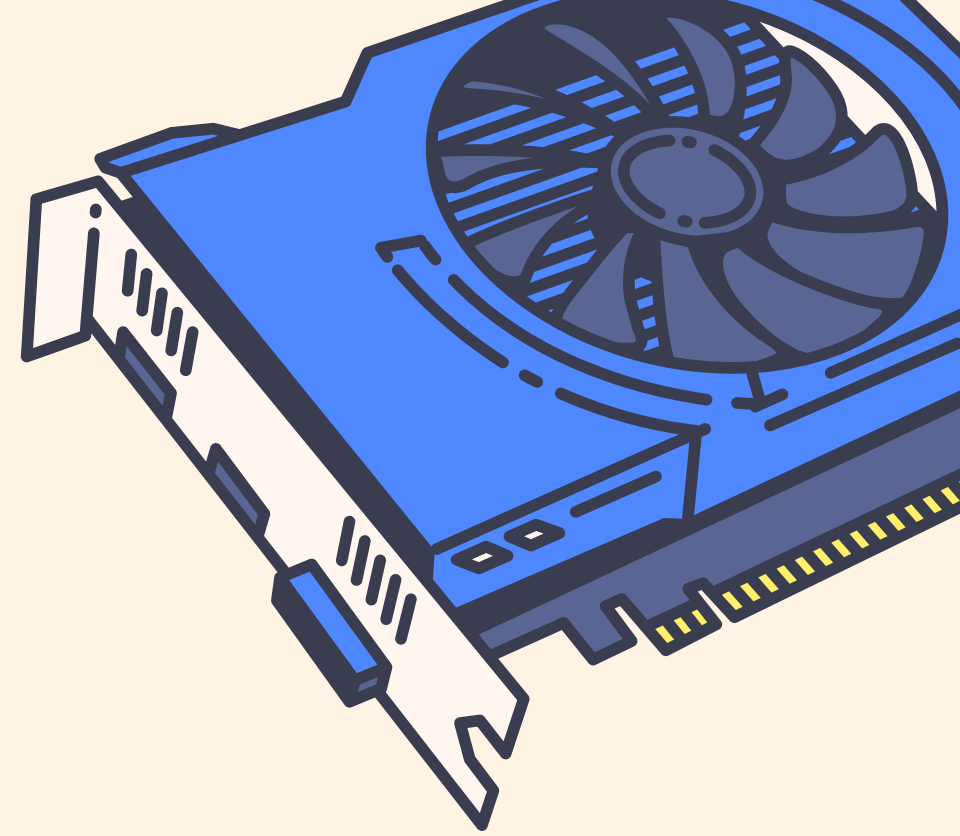
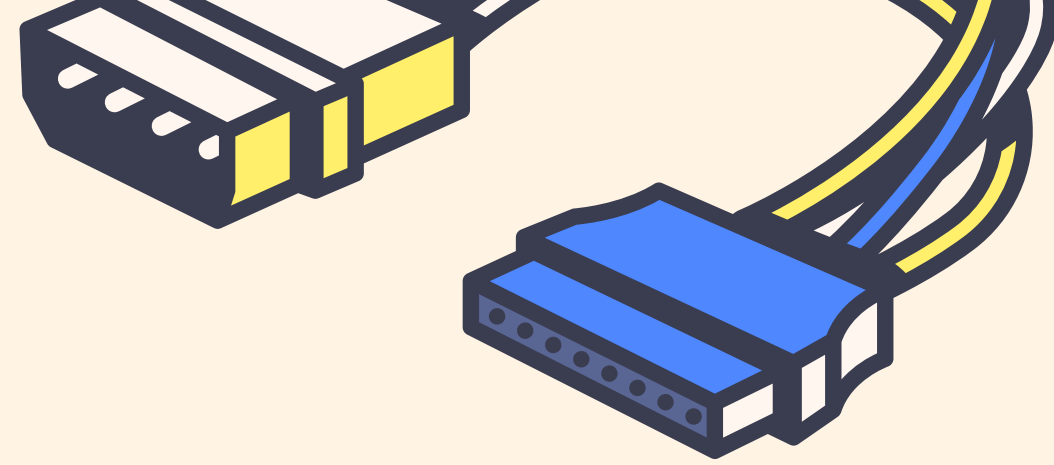
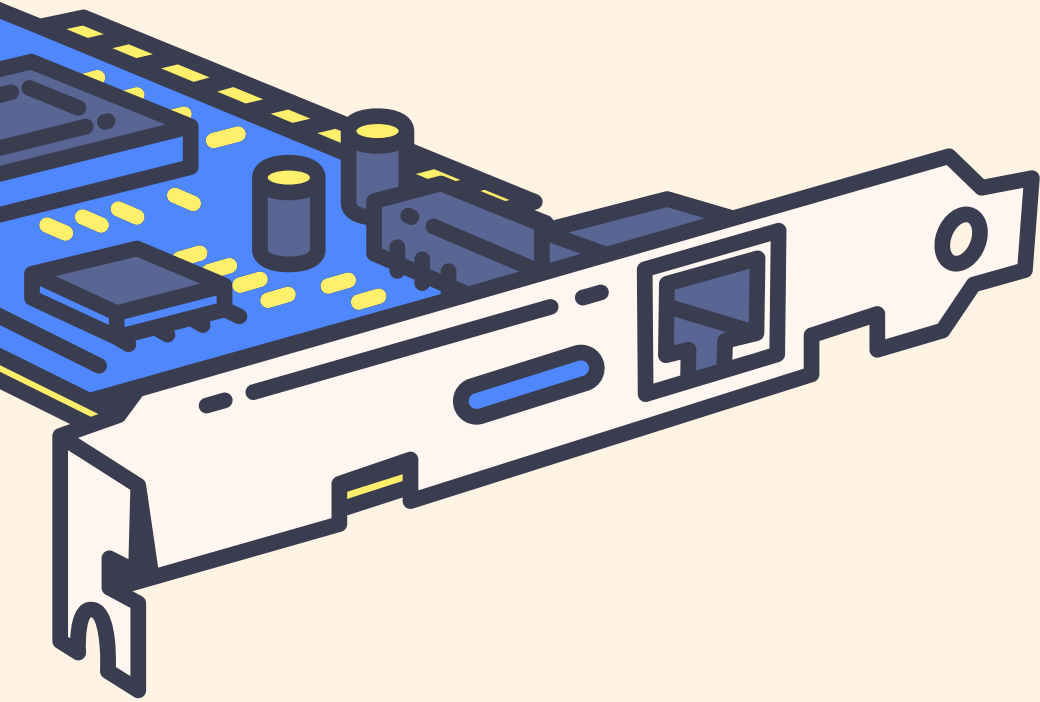
Request 10 :

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

```
with cte1 as
(
SELECT
    p.product_code,
    p.product,
    p.division,
    SUM(s.sold_quantity) AS total_sold_qty
FROM
    gdb023.dim_product p
    JOIN
    fact_sales_monthly s ON p.product_code = s.product_code
WHERE
    s.fiscal_year = 2021
GROUP BY product , s.product_code , division
),
cte2 AS (
SELECT * , RANK() OVER(PARTITION BY division ORDER BY total_sold_qty desc ) AS rnk
FROM cte1)
SELECT * FROM cte2
WHERE rnk IN(1,2,3) ;
```

product_code	product	division	total_sold_qty	rnk
A6720160103	AQ Pen Drive 2 IN 1	N & S	701373	1
A6818160202	AQ Pen Drive DRC	N & S	688003	2
A6819160203	AQ Pen Drive DRC	N & S	676245	3
A2319150302	AQ Gamers Ms	P & A	428498	1
A2520150501	AQ Maxima Ms	P & A	419865	2
A2520150504	AQ Maxima Ms	P & A	419471	3
A4218110202	AQ Digit	PC	17434	1
A4319110306	AQ Velocity	PC	17280	2
A4218110208	AQ Digit	PC	17275	3





Thank You

